AI Energy Demand and the Economic Case for Space Data Centers

A transparent, interactive model: methodology and assumptions

Prepared by Claude Opus 4.8 (Anthropic) under the supervision of Alessandro Golkar, Technical University of Munich, golkar@tum.de

Technical report v1.0 · 2026 · companion to the interactive model at golkar.ai

[!] AI-generated, under verification. This model and report were generated by Claude Opus 4.8 (Anthropic) under the supervision of A. Golkar and are currently undergoing verification. Do not trust the results unless manually verified. Equations below describe what the code computes; they are not independently peer-reviewed.

1. Purpose and scope

The model estimates future electricity demand from artificial intelligence, compares it to the electricity the Earth can supply, identifies the point at which demand outruns supply, and evaluates whether the resulting gap could be served more cheaply from orbit than from additional ground buildout. It is deliberately transparent: every assumption is an adjustable parameter, and the simulation is a pure function of those parameters. The aim is a tool for structured reasoning under deep uncertainty, not a point forecast.

2. Model architecture

Demand is built bottom-up from three segments evaluated annually from a base year to a horizon year: (i) consumers, (ii) corporate knowledge workers, and (iii) autonomous agents. Each segment converts a population of users into electricity via a per-user query rate and a per-query energy. Supply is global electricity generation growing at a chosen rate; AI demand is stacked on top of non-AI demand (homes, industry, transport) and the total is compared against generation. A breakeven year and an Earth-versus-space cost comparison are derived from the resulting trajectories.

3. Demand model

3.1 Logistic adoption

Each human-bounded segment follows a logistic (S-curve) from a starting level a_0 toward a ceiling L at rate r, in years t after the base year:

```
A(t) = L / (1 + exp(-r (t - t0))),   t0 = ln((L - a0)/a0) / r
```

3.2 Population base

Population P and the knowledge-worker base grow geometrically at rate g:

```
P(t) = P0 (1 + g)^t
```

3.3 Consumer and corporate segments

Consumer users are the population times consumer adoption; corporate users are the knowledge-worker base times corporate adoption. Both are bounded by human headcount, so both saturate.

```
N_consumer(t) = P(t) · A_consumer(t)
N_corp(t) = W(t) · A_corp(t)
```

3.4 Autonomous agents (decoupled from headcount)

Agents are modelled as a compute-driven fleet whose saturation is expressed as agents-per-human against the base population. This is a transparent sizing of the ceiling; the fleet can far exceed the number of people. The fleet itself follows a logistic toward that ceiling:

```
C_agents = (agents per human) · P0
N_primary(t) = logistic(fleet0, C_agents, r_agent, t)
```

Recursive sub-agents multiply the live fleet. If each agent spawns Z sub-agents over D generations, the total instance count is the geometric series 1 + Z + Z^2 + ... + Z^D:

```
M(Z, D) = (Z^(D+1) - 1) / (Z - 1)  (M = D+1 when Z = 1)
N_agents(t) = N_primary(t) · M(Z, D)
```

M grows exponentially in depth D, so this term can dominate all others. It is disabled by default (D = 0, hence M = 1) so the central case is not inflated.

3.5 Energy intensity: efficiency versus Jevons rebound

Per-query energy evolves under two opposing forces. Hardware and algorithms reduce energy per query at rate e (negative); a Jevons rebound R re-spends a fraction of those savings on more and heavier work. The net trend applied to the base energy w_0 is:

```
net = (e/100) · (1 - R/100)
w(t) = w0 (1 + net)^t
```

At R = 100% the forces cancel (net-flat per-query energy); above 100% demand outpaces efficiency; below 100% efficiency lowers demand over time.

3.6 Segment energy

For a segment with N (billions of) units, q queries per unit per day, per-query energy w (Wh), agent intensity multiple m (1 for humans), facility PUE, and training overhead h, annual energy in TWh is:

```
E = N·10^9 · q · 365 · w · m · PUE · (1 + h/100) / 10^12
D(t) = E_consumer + E_corp + E_agent
```

4. Supply and the demand stack

Global generation S grows geometrically. Electricity is consumed as it is generated, so the grid is essentially fully used today; there is no idle surplus waiting for AI. The model therefore tracks non-AI demand B (homes, industry, transport) with its own growth rate, and treats AI demand D as stacked on top of it. Total electricity demand is the sum, compared against generation:

```
S(t) = S0 (1 + gs)^t  (generation)
B(t) = B0 (1 + gp)^t (1 + gc)^t  (non-AI demand)
D_total(t) = B(t) + D(t)  (non-AI + AI)
```

Non-AI demand B scales with both population growth gp and a per-capita electrification trend gc, so it responds to the population assumption rather than drifting independently. It is set close to current generation at the base year, so AI begins as a thin sliver on top of an almost-full grid. AI can grow only into the margin that generation opens over non-AI demand. If non-AI demand grows as fast as or faster than generation, that margin never opens and AI can grow only by displacing other uses.

5. Breakeven definition

The breakeven is the first year total demand (non-AI plus AI) exceeds generation: the point at which the world cannot produce enough electricity to power both the rest of the economy and AI. It is the physically grounded signal that new capacity is needed.

```
t* = min { t : B(t) + D(t) > S(t) }
```

6. Earth versus space economics

The unserved gap is total demand beyond generation. Capacity is sized against the largest gap over the horizon (efficiency trends can make demand peak then ease, so the final year is not always the binding year). The gap energy is converted to an average continuous power:

```
gap(t) = max(0, B(t) + D(t) - S(t))
W_gap = gap_peak · 10^12 / 8760  [W]
```

Closing it on Earth costs the gap power times an all-in build cost per watt. In orbit, generation mass scales down with the orbital solar advantage (a panel in the right orbit is several times more productive), while compute and bus mass (~half) do not; the mass is priced via the launch cost and a fixed non-launch hardware cost:

```
Earth$ = W_gap · cost_earth
kg/kW_eff = kgPerKw / solarMult + 0.5 · kgPerKw
Space$/W = (kg/kW_eff / 1000) · launch$/kg + nonLaunch$/W
Space$ = W_gap · Space$/W
```

Setting the two equal and solving for launch price gives the threshold at which orbit becomes cheaper than the ground, the number the business case hinges on:

```
launch*_$/kg = (cost_earth - nonLaunch$/W) / (kg/kW_eff / 1000)
```

7. Parameters and default values

Defaults are seeded from public data where it exists (mid-2026) and from reasoned estimates otherwise. Estimates are flagged. The agents-per-human value and recursion depth are the dominant swing variables and are deeply uncertain; treat them as one point in a wide range, not a forecast.

Parameter	Default	Basis / source
Population & time		
World population	8.3 B	Worldometer / UN, mid-2026
Population growth	0.8 %/yr	UN WPP; held constant
Horizon	2026-2050	Modelling choice
Agents per human (ceiling)	7.5 /person	Anchor ~50 per knowledge worker; swing variable, est.
Consumer		
Adoption today / ceiling	0.12 / 0.90	~1B of 8.3B use a chatbot (OpenAI, Feb 2026)
Adoption speed	0.35 /yr	Estimate
Queries / user / day	12	Estimate from OpenAI usage disclosures
Corporate		
Knowledge workers	1.25 B	~1.25B of ~3.5B workforce; estimate
Adoption today / ceiling	0.10 / 0.85	Deep per-worker augmentation ~10%; surveys
Adoption speed	0.40 /yr	Estimate
Queries / worker / day	60	Estimate
Autonomous agents		
Fleet today	0.05 B	Estimate; nascent
Fleet growth	0.45 /yr	Gartner/IDC near-term agent adoption; est.
Queries / agent / day	400	Estimate; 24/7 operation
Agent energy multiple	10x	Reasoning/agentic order of magnitude (Epoch AI)
Sub-agents per agent (Z)	3	Speculative; only active when D > 0
Recursion depth (D)	0	Off by default; exponential wild-card
Energy intensity		
Energy per query	0.30 Wh	Median optimized text query (Epoch AI, IEEE)
Facility PUE	1.25	Uptime Institute hyperscale
Training overhead	25 %	Estimate (10-40% range)
Hardware efficiency	-20 %/yr	AI compute efficiency trend (Epoch AI)
Jevons rebound	100 %	Neutral midpoint (net-flat per query)
Supply & non-AI demand		
Global generation	30,664 TWh	Ember Global Electricity Review 2025 (2024)
Supply growth	3.0 %/yr	IEA (4% in 2024; 2.6% 2010-23 avg)
Non-AI demand today	30,600 TWh	~ current generation; AI is ~0.1% today
Non-AI per-capita growth	1.2 %/yr	Electrification; total adds population growth (~2%/yr total)
Earth vs space		
Earth build cost	3.0 $/W	Estimate, generation + data center
Launch cost	1,000 $/kg	~1,500-2,900 today; targets 200-500 (Sci.Am./DCD)
Mass per kW (space)	6 kg/kW	Estimate, panels+radiators+bus+chips
Orbital solar gain	8x	Google Project Suncatcher

Parameter	Default	Basis / source
Space hardware cost	2.0 $/W	Estimate, excl. launch

8. Key findings (default scenario)

Under the default assumptions, AI electricity demand rises from about 0.1% of global generation in 2026 to roughly 0.6% in 2030, 28% in 2040 and 67% in 2050. Because non-AI demand already consumes almost the entire grid, total demand (non-AI plus AI) crosses generation in 2037, after which an unserved gap opens, reaching on the order of 28,000 TWh (a few thousand GW of continuous power) by 2050. The agent segment dominates AI demand; consumer and corporate use are small by comparison. Space becomes cheaper than additional Earth buildout once launch cost falls below roughly 270 $/kg, consistent with the published Starcloud (~500 $/kg) and Google Suncatcher (<200 $/kg) targets.

Robust vs contingent. Robust across essentially the full plausible range of the agent ceiling: total demand outruns generation in the late 2030s, opening a multi-thousand-gigawatt gap. Contingent and genuinely uncertain: whether AI alone ever exceeds total global generation, which is governed almost entirely by agents-per-human and the recursion depth. The model affirms the first and leaves the second explicitly open.

9. Limitations and caveats

(1) Supply is a smooth exponential and ignores grid build-out friction, siting, and intermittency. (2) Demand saturation is imposed through logistic ceilings and a fixed agent ceiling; the true ceiling is economic and unknown. (3) Sub-agent recursion is a simple geometric tree; real sub-agents differ in lifetime and load. (4) Per-query energy, query rates, and the agent intensity multiple are estimates with wide uncertainty. (5) The space cost model is a first-order capex comparison and omits operations, cooling at scale, radiation hardening, latency, debris, and regulatory cost. (6) All figures are AI-generated and pending manual verification. The model's value is comparative and exploratory, not predictive.

10. Selected sources

- Ember, Global Electricity Review 2025 (2024 generation 30,664 TWh).
- IEA, Global Energy Review 2025 and Energy & AI 2025 (supply growth; data-center demand to 2030).
- Epoch AI, energy per query (~0.3 Wh median); IEEE Spectrum, AI energy use.
- OpenAI / TechCrunch (Feb 2026), ChatGPT ~900M weekly users.
- Gartner / IDC, enterprise agentic-AI adoption forecasts (direction only).
- Scientific American; Data Center Dynamics, space data centers (Starcloud, Google Project Suncatcher); launch-cost targets.